

HTML Learning Tasks

Task 1: Semantic vs Non-Semantic HTML Elements

Objective

Understand the difference between semantic and non-semantic HTML elements and learn when to use each type appropriately.

Topics Covered

Semantic Elements

- `<header>` - Defines introductory content or navigation links
- `<main>` - Represents the main content of the document
- `<footer>` - Contains footer information like copyright, contact details
- `<nav>` - Navigation section with `aria-labelledby` for accessibility
- `<article>` - Self-contained, independent content
- `<aside>` - Content tangentially related to the main content
- `<details>` and `<summary>` - Interactive disclosure widget
- `<time>` - Represents dates/times with `datetime` attribute

Non-Semantic Elements

- `<div>` - Generic container for block-level content
- `<span>` - Generic container for inline content

Practical Exercise

Create a complete webpage structure using semantic HTML elements. Your page should include:

1. A header with site navigation using `<nav>` and `aria-labelledby`
2. A main content area with at least two articles
3. An aside section with related information
4. A details/summary element for expandable content
5. At least one time element with proper `datetime` attribute
6. A footer with contact information
7. Use `div` and `span` only where semantic elements don't apply

Success Criteria

- Proper nesting of semantic elements
 - Correct use of aria-labelledby on navigation
 - Valid datetime attributes on time elements
 - Clear distinction between when to use semantic vs non-semantic elements
-

Task 2: HTML Tables - Structure and Accessibility

Objective

Master the creation of well-structured, accessible HTML tables with proper organization and semantic meaning.

Topics Covered

Basic Table Structure

- `<table>` - Container for tabular data
- `<tr>` - Table row
- `<th>` - Table header cell
- `<td>` - Table data cell

Advanced Table Features

- `rowspan` - Span cell across multiple rows
- `colspan` - Span cell across multiple columns
- `<caption>` - Table title/description
- `<thead>` - Groups header content
- `<tbody>` - Groups body content
- `<tfoot>` - Groups footer content
- `scope` attribute - Defines header cell scope for accessibility

Practical Exercise

Create three different tables:

1. **Simple Data Table:** A basic 4x5 table showing product information with proper headers
2. **Complex Table with Spanning:** A schedule/timetable using `rowspan` and `colspan` to merge cells

3. **Fully Accessible Table:** Include all structural elements (caption, thead, tbody, tfoot) with proper scope attributes

Success Criteria

- Proper use of thead, tbody, and tfoot sections
 - Correct implementation of rowspan and colspan
 - Descriptive captions for all tables
 - Appropriate scope attributes (row/col/rowgroup/colgroup) on header cells
 - Semantic structure that makes sense when read by screen readers
-

Task 3: Complete HTML Fundamentals

Objective

Build a solid foundation in HTML by mastering all essential elements and creating a complete, professional webpage.

Topics Covered

1. HTML Introduction

- Understanding HTML structure and purpose
- Document type declaration and basic syntax

2. VS Code Setup

- Installing Visual Studio Code
- Essential extensions for HTML development
- Using Emmet shortcuts and live preview

3. Basic HTML Structure

- DOCTYPE, html, head, and body tags
- Proper document structure

4. Meta Information

- `<meta name="author">` - Page author
- `<meta name="description">` - Page description for SEO
- `<link rel="icon" type="image/x-icon">` - Favicon

5. Text Basics

- `<h1>` to `<h6>` - Headings hierarchy
- `<p>` - Paragraphs
- `<hr>` - Horizontal rule/thematic break
- `<br>` - Line break
- `<strong>` - Strong importance (bold)
- `<em>` - Emphasis (italic)
- HTML Entities: ` `, `<`, `>`, `©`, etc.
- `<abbr>` - Abbreviations with title attribute
- `<address>` - Contact information

6. Lists

- `<ol>` - Ordered (numbered) lists
- `<ul>` - Unordered (bulleted) lists
- `<dl>`, `<dt>`, `<dd>` - Description lists

7. Links

- `<a href="">` - Hyperlinks with relative and absolute paths
- Internal links using id attributes
- `<section id="">` - Creating anchor points
- `<nav>` - Navigation container
- download attribute - Force file download
- `href="mailto:"` - Email links
- `href="tel:"` - Phone links
- `target="_blank"` - Open in new tab

8. Images

- `<img>` with attributes: `src`, `alt`, `width`, `height`, `title`, `loading`
- `<figure>` and `<figcaption>` - Semantic image containers

9. Semantic HTML

- Complete semantic structure (header, main, footer, nav, article, aside)
- `aria-labelledby` for enhanced accessibility

- <details> and <summary> for interactive content
- <time datetime=""> - Dates and times

10. Non-Semantic Elements

- <div> - Block-level container
- - Inline container

11. Tables

- Full table structure with accessibility features
- Spanning cells and proper organization

Final Project

Create a complete multi-page website that includes:

1. Home page with proper meta tags, semantic structure
2. About page with text formatting, lists, and contact information
3. Gallery page with images using figure/figcaption
4. Schedule/data page with a complex table
5. Navigation system linking all pages
6. Proper use of semantic vs non-semantic elements throughout

Success Criteria

- Valid HTML5 structure on all pages
 - Proper SEO meta tags
 - Accessible navigation with aria-labelledby
 - Correct use of all text formatting elements
 - Various list types implemented appropriately
 - All link types demonstrated (internal, external, email, tel, download)
 - Images with proper attributes and semantic containers
 - At least one complex table with full structure
 - Consistent semantic HTML throughout
 - Clean code with proper indentation and comments
-

Assessment Tips

- Validate your HTML using W3C Markup Validation Service
- Test accessibility with browser developer tools
- Ensure your code is properly indented and readable
- Add comments to explain complex sections
- Test all links and ensure images load correctly